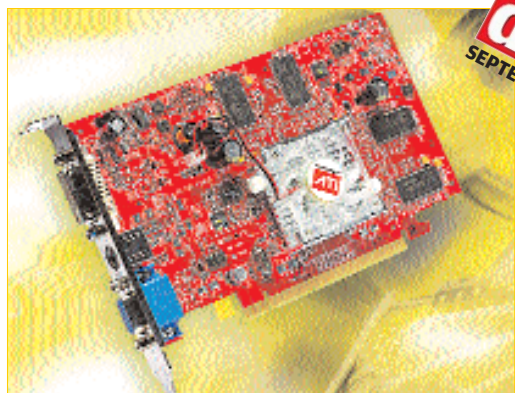




BEST BUY
digit
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PowerColor X300
128 MB

Whatever it won in terms of pure performance and features, the PowerColor X300 256 MB card lost when considering its price of Rs 5,950. It still managed to scrape in as our Silver winner.

MID-RANGE CARDS

The mid-range cards always hold the interest of those who want the best, but can't afford it. Such users usually settle for the next-best. We got plenty of cards to play with this time—ten cards from nVidia and three from ATi. We could not include one card from Asus, the 6600GT, since it suffered from display corruption. We also had two new entrants in this category, namely BIG and Sapphire, who sent us nVidia and ATi cards respectively.

Features

Most cards in this category were bundled with the regular assortment of cables and games, but some brands stood out. Sapphire, for instance, bundled a VIVO cable with their card, while it

rained games in the MSI box—we counted 14 CDs, including game and software application CDs. As usual, Gigabyte, too, provided an excellent game and software bundle with their cards.

The rest of the cards were modestly packaged. However, the performance of the cards was what we were waiting for, and that's what we talk about next.

Almost all the cards we tested in this category were based on either the 6600 or 6600GT chips from nVidia, or the X600 or 600XT chips from ATi. Before we move on to performance, let's look at some differences in the cores of these two chips. This will let us better analyse the performance difference that will be apparent when you check the scores.

The 6600GT core has been built from the ground up for the PCIe bus and runs at a core clock speed of 500 MHz, which is actually faster than a 7800GTX. It also features eight pixel pipelines, which puts it in the same class as the Radeon 9700.



Gigabyte GeForce 6600
TurboForce Edition 256MB

How We Tested

The low-end segment of our tests comprised eight cards based on the nVidia GeForce 6200TC and the ATi X300 chips. The mid-range segment comprised thirteen cards based on the nVidia GeForce 6600 series and the ATi X600 and X700 series chips, while nine cards based on the nVidia 6800 series and the 7800 GTX and ATi X8xx series chips were included in the high-end segment. We also tested the MSI NX6800GT SLI and the ASUS Dual GPU GeForce 6800 Ultra cards separately.

We had two different test beds, one for the low-end and other for the mid-range and high-end cards. The test bed for the low end comprised of a 939-pin AMD Athlon64 3800+ on an ECS RS480-M motherboard with 512 MB of 400 MHz DDR Corsair TwinX RAM modules and a 120 GB Seagate SATA hard drive.

The other test bed comprised an AMD Athlon64 FX-53 on an nForce4 SLI edition-based motherboard with high performance and low latency 1 GB of 400 MHz DDR Corsair XMS TwinX RAM modules, and a 250 GB Maxtor SATA hard drive. The test beds above were chosen after careful consideration to eliminate any CPU and memory bottlenecks in the respective categories. All the tests were performed at 32-bit colour depth.

Low-end card tests: The testing process for the different categories was also different. In the low-end segment, we benched *Doom 3*, *Half-Life 2* and *FarCry* at lower resolutions of 640 x 480, 800 x 600 and 1,024 x 768, and the frames per second were logged. *Doom 3* tested the OpenGL graphics subsystem, while *Half-Life 2* and *FarCry* tested the Direct3D subsystem. Futuremark's popular synthetic benchmark, 3DMark 03, was also run at 800 x 600 and

1024 x 768 at default settings, and the score was noted. During the tests, anti-aliasing (AA) and anisotropic filtering (AF) was turned off.

Mid-range and high-end card tests: For the mid-range and high-end cards, we used the above-mentioned games as well as *Halo* and Futuremark's 3DMark 05. All the gaming tests were performed at 1024 x 768 with AA and AF turned off, and the scores were noted. *Doom 3* was tested at 1,024 x 768 at high quality and 4x AA settings. It was also tested at 1,280 x 1,024 and 1,600 x 1,200 at high quality settings with AA turned off. *Half-Life 2*, *FarCry* and *Halo* were tested at high quality settings at 1,024 x 768, 1,280 x 1,024 and 1,600 x 1,200 at high settings, and at low-quality settings at 1,024 x 768.

3DMark 03 and 3DMark 05 were run at default settings at 1,024 x 768 and the scores were noted. In all the cases, AA and AF were turned off. The MSI NX6800GT SLI and the Asus Dual GeForce 6800 Ultra GPU cards were tested the same way.

How We Awarded

For the low-end products, price was given a considerably higher weightage than the other categories. The price weightage was reduced for mid-range and high-end cards in favour of weightage for performance and features.

The performance weightage for the high-end cards was the highest. The scores were calculated on the basis of features, performance and price.

The card with the highest score was awarded the Digit Best Buy Gold award and the first runner-up was awarded the Digit Best Buy Silver award.